

# Process Performance Measurement

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**Abstract** Measuring performance is a precondition for analyzing and subsequently improving business processes. However, selecting the “right” criteria for measurements remains a difficult and highly debated topic, both in theory and practice. The objective of this chapter is to provide an overview how adequate measures can be identified and implemented. For this purpose, the main terms will be defined and the procedure of process measurement will be presented. This includes a number of approaches for measuring process performance. In more detail, two methodologies, Data Envelopment Analysis and Process Mining, to measure and calculate indicators for process performance are discussed. As an example, Process Mining is applied to real case data to demonstrate its possibilities.

## 1 Measurement as Part of Process Management

Managers become more and more aware of the necessity to design and to control companies from a process-oriented perspective. As a consequence, the entire company should be strictly aligned to the requirements of its customers. The management of the necessary processes should follow the typical business process management cycle (van der Aalst et al. 2003). Thus, each business process (in the following the term ‘processes’ is used) should be designed, implemented, executed, and controlled. Process control includes the measurement, analysis, and improvement of processes. In this chapter we focus on the first step of process control, that is to conduct accurate measurements (Ljunberg 2002).

Process measurement is the continuous observation of predetermined performance indicators for the purpose of attaining process targets (Kueng and Kawalek 1997). Although measuring a company’s process performance is a topic that has

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been broadly discussed in publications, precise definitions of performance measurement have been rarely provided.<sup>1</sup> According to Neely et al. (2005, p. 1229) "... the level of performance a business attains [is] a function of the efficiency and effectiveness of the actions it undertakes." In their view, measurements are defined and established in order to define performance. A coordinated and aligned set of measurements (metrics) represents a measurement system that is suited to quantifying efficiency and effectiveness (performance).

In order to measure processes four generic steps have to be considered (Ljunberg 2002; Neely et al. 2005; Yen 2009). Firstly, it has to be determined how the measurement in general should be performed. For this purpose, several frameworks are available describing the set-up of a process measurement system. Secondly, the indicators for measuring process performance have to be identified. Thirdly, the data sources to gather the necessary data have to be selected. Fourthly, the measurements should consider implications from the context of the respective business process.

The objective of this book chapter is to provide an overview how the performance of processes can be measured. Therefore, in Sect. 2 the term performance of business processes is clarified. Section 3 presents the procedure of process measurement which covers the steps from identifying performance indicators to the consideration of the context. Within Sect. 4 two prominent methodologies for the measurement of process performance, Data Envelopment Analysis and Process Mining, are presented. In Sect. 5, the latter methodology is illustrated in detail by means of an application on real case data. Section 6 concludes the chapter by summing up the discussed topics.

## 2 Defining Process Performance

Performance of business processes is about how well the executed processes work with regard to the chosen indicators. Performance can be seen from two perspectives (Coelli et al. 2005):

- Descriptive measures, i.e., how the performance of the process is actually observed. The chosen indicators of a process are measured as the relation of output to input. An aggregation is required if more than one input or output is considered. This perspective is termed as **productivity**.
- Normative measures, i.e., how the performance of the process should be. Either the actual output is compared with the maximum producible quantity of output from its observed input (output-oriented) or the actual input with the minimum required input for its observed output (input-oriented). This perspective is termed as **efficiency**. Another normative indicator is **effectiveness** which is focusing on the question if a process is focusing on the adequate output at all. "It is fundamentally

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<sup>1</sup> Also see vom Brocke and Sonnenberg (2014) in this handbook, who provide a discussion and conceptualization for performance evaluations in BPM.

the confusion between effectiveness and efficiency that stands between doing the right things and doing things right. There is surely nothing quite so useless as doing with great efficiency what should not be done at all” (Drucker 2006, p. 145).

The consequence of the two perspectives is that the measurement of productivity is a precondition for the identification of process efficiency and process effectiveness. Every indicator of a process is the result of transforming an input into an output. For the definition of inputs and outputs, there are various possibilities, such as costs per employee, working time for an activity or number of sales orders processed (Johnston and Jones 2004).

A stand-alone productivity indicator has no reference to the determined value. Only by a comparison with other productivities or other reference values, meaningfulness is created. Various productivity indicators can be compared or be placed in a sequence to each other. If the productivity ratios are comparable in their design, it is also possible to calculate the relative distance from the reference values. As a constant ratio of input to output over time should be used, possible economies of scale in a comparison of productivity cannot be identified (McLaughlin and Coffey 1990).

Process performance measurement entails capturing quantitative and qualitative information about the process. Subsequently, the measurements can be transformed into performance figures which translate unfiltered data into information about process performance, enabling the process manager to deal with process control.

### 3 Procedure of Measuring

#### 3.1 Design of a Process Measurement System

Process performance is multi-dimensional, i.e., process performance cannot be determined on the basis of a single indicator, such as productivity, but results from many different indicators, measures, and performance figures. In addition, process performance indicators are not independent from each other. Most process performance indicators exhibit a relationship with other indicators, i.e., they complement one another or they conflict with each other (Gillies 1995).

As a result, a multitude of frameworks for determining a performance measurement system exists of which important ones will be discussed in the following. The framework developed by Fitzgerald et al. (1991) represents the insights gained from research in service industries. The authors divide performance indicators into two basic types of *Performance Indicators*:

- indicators that are related to and reflect process results (measurements assessing the competitive environment, such as market share or market growth, as well as financial measurements, such as costs and profits) and
- indicators that reflect the determinants of process success (measurements to determine quality, flexibility, resource utilization, and innovation).

Another well-known framework for designing a measurement system is the *Performance Pyramid* (Lynch and Cross 1991). This framework follows a hierarchical view of performance. It considers the relationship between strategic performance (e.g., fulfilling the vision, market share, financial performance) and process performance (e.g., quality, cycle time, waste, or spoilage rate). The layer connecting the two hierarchical levels depicts those performance indicators that impact both levels (e.g., customer satisfaction, flexibility, and productivity).

A stronger process perspective is emphasized by Brown (1996). His framework highlights the distinction between input, throughput, and output as he recommends determining performance indicators according to this classification. Process input factors include, for example, employees, plant and equipment, as well as capital. The quality and quantity of these input factors can be a decisive factor in meeting customer requirements which themselves represent an additional input factor. During the throughput phase, input factors are utilized and combined. The output consists of a product, a service, and financial results. The process performance is therefore determined through a measurement system that encompasses performance measures for input, throughput, and output. In addition, performance measures related to meeting customer requirements and customer satisfaction are included.

Kaplan and Norton (1992) suggest eight steps, which they believe managers should follow in order to develop a measurement system in alignment with the Balanced Scorecard (Kaplan and Norton 1993): (1) prepare measurement system development, (2) conduct one-on-one interviews with senior managers (to elicit different vision and strategy approaches), (3) hold a first workshop with senior management (to agree on vision and strategy), (4) conduct extended interviews with senior management (to identify different indicators), (5) hold a second workshop with senior management (to determine performance indicators and target measures for each indicator), (6) conduct a third workshop with senior management (to define the Balanced Scorecard), (7) implement the measurement system, and (8) periodically review the measurement system. This procedure highlights how the measurement system should be aligned with vision, strategy, and strategic objectives.

Setting up a measurement system is a sensible undertaking. The outlined frameworks provide managers with guidance with respect to the development of an individualized measurement system. In doing so, emphasis should be placed on championing a methodology suited to developing a measurement system rather than prescribing a general “one-size-fits-all” system. To develop an individualized process performance measurement system, implementation steps have to be taken. These can be based on the recommendations by Brignall and Ballantine (1996), Fitzgerald et al. (1991), and/or Kaplan and Norton (1993). Subsequently, the external environment, the strategy, and the process model of the company have to be taken into account. An example of such a dynamic methodology is the framework delivered by Neely et al. (2000).

### 3.2 Identification of Performance Indicators

Measuring process performance starts with the identification of performance indicators that allow for a detailed specification of process performance. Therefore numerous authors have suggested categories of indicators in order to facilitate a structured approach. The majority of authors, such as Sarin et al. (2011) as well as Wu and Liu (2008), have adopted a process-oriented view, resulting in the indicator groups quality, time, costs, and flexibility. These four indicator groups will be examined in the following.

In general, *quality* describes the degree to which the actual product attributes and properties conform to the underlying product specifications. In the past, indicators often included costs, for example, defect prevention, quality measurement costs, and costs related to failure rates (Campanella and Corcoran 1983). However, the emergence of quality improvement initiatives such as Total Quality Management has resulted in a new definition of the term quality. Nowadays, customer satisfaction serves as the yardstick for measuring the quality of a product or service.

From a production point of view, *time* is considered to be an indicator for competitiveness and process performance. Yet, the time aspect can be looked at from different angles: Within a just-in-time production paradigm, for example, production and/or delivery of production outputs at a premature or belated point of time is considered to be a waste of time. Thus, the exact point of time is relevant, and an appropriate performance indicator would be the deviation from the targeted point of time. Within the field of research in Optimized Production Technology (OPT), the main objective is seen in minimizing process time (McLaughlin and Coffey 1990). Performance indicators and subsequent performance measures therefore include, for example, throughput time, actual processing time, waiting time, transportation time, and delivery time.

Research in the area of cost accounting provides the basis for the determination of *cost* indicators (e.g. Ravichandram 2007). A large number of academic contributions, for example Johnson (1983), deal with the various cost aspects. Different cost factors provide the basis for cost indicators: labor costs, IT costs, production costs, product costs, service costs, failure costs, and so forth. A distinction can be made between fixed and variable costs. Since the emergence of activity-based costing, indicators such as activity-based costs, sub-process-, or process-related costs are also feasible.

Indicators to determine *flexibility* include, according to Slack (1987), the degree to which a production or service process can be modified, including the timeline and costs associated with the restructuring of a production or service process. A further indicator for flexibility relates to the number of product or service components that can be exchanged within a given time. Moreover, process flexibility may also be viewed as dealing with output volumes or resource utilization.

In summary, there are many indicators available that can be applied for the measurement of company-specific process performance. Companies have to select indicators that are directly linked to their strategy, and they have to link the

indicators to their business objectives and resources. This will then result in strategic performance figures that support senior management in navigating toward the desired strategic direction. In order to be successful, each company has to determine performance indicators and, subsequently, performance measures and performance figures that are strategically relevant to its respective situation.

### 3.3 *Determination of Data Sources for Measurement*

Several techniques can be used to obtain the relevant information using different data sources. Sometimes the techniques are used independently from each other, but mostly they are used in combination during the entire survey period. Each collection technique allows obtaining a certain amount of information. Therefore, it is particularly challenging to determine the right mix of techniques to gather relevant data.

*Studying documents* is often the first step of data collection. The relevant information can be found in customer and order documents, job descriptions, organizational manuals, work instructions, databases, and current literature. A major goal of the document study is to obtain an overview of the quality and quantity of the existing documentation of processes. Furthermore, a basic understanding of the technical and organizational contexts can be achieved. Thus, studying documents is a helpful data source for identifying further needed data sources (Ungan 2006).

In most cases *interviews* with employees of the respective organizational unit are conducted. The interview is the most open and flexible format to gather data, as the questioner is able to control the conversation and thereby the flow of information by using appropriate questioning techniques. It is usually distinguished between a free and a structured interview. During the conversation, it is possible to clarify traditional practices during process execution, reservations to changes and informal communication structures (Johnson 2002).

*Observation* is an optical recording and interpretation of the activities of the monitored person. Observations can be taken officially or unknown to employees. A significant advantage of observation with respect to interviews is the independence from the ability and willingness to provide information of the observed person. A special type of observation is the *multi moment study*. Here, a sample of individual observations is made in regular intervals. This sample is generalized to the whole working area using statistical methods (Castle and Harvey 2009).

Within *questionnaires*, a list of questions will be presented to employees. The advantage is that at the same time a large number of people can be addressed. It must however be noted that manipulation through conscious or unconscious false statements is easy to be made, but difficult to detect. Due to the missing interaction between interviewer and interviewee, there is no possibility to intervene directly in

the response process. This complicates the collection of clear and focused information (Yen 2009).

Especially for the determination of a job or position profile and the determination of processing times *self-recording* can be used. In this case, the collection of data is transferred to the employees involved in each study area. The employee notes all activities and times in a special document or collection system during his/her daily work.

*Order worksheets* are attached to each object (e.g., document or parcel) which is processed. For each step performed information is added on the worksheet typically including description, originator, and time. This technique is particularly suitable for determination of working, idle and transportation time. Order worksheets can also be created automatically by information systems in form of event logs which are described in Sect. 4.2.

*Estimations* are used for approximate determination of data. The estimations are often based on past values which are updated based on statistical methods (e.g., trend extrapolation). Usually estimations are conducted by experts in the related work area.

The choice of the data source depends on the purpose but also on availability. In practice, often data sources are not available because of missing information systems or obtaining the needed data is too complex.

### 3.4 Contextual Implications of Measurement

Identifying the root causes for process performance deficiencies and eliminating them is an important task of every company. In doing so, it should be considered that the context in which a business process is executed plays an important role for variances in process performance (Banker and Natarajan 2008).

The influence of the context can be operationalised by contextual factors. These are indicators of the environment surrounding a business process. Contextual factors can be found outside of the external corporate environment or within internal company characteristics (Papadakis et al. 1998). In any case, they are not inherent in the business process under analysis. Examples of contextual factors are weather conditions or holiday seasons (Rosemann et al. 2008).

Within process operations, the consideration of contextual factors could lead to a diversified view of measuring process performance. Instead of analyzing the performance of a business process as a whole, it might be favorable to derive different performances based on the major independent contextual factors. Processes should be improved having a multi-performance in mind. One specific characteristic of a contextual factor might be the reason for a poor overall performance. Using a call center as an example, the root cause of low performance could be the time of the day, e.g., the evening. The reason for this might be a peak of calls caused by

customers calling after work while the number of employees working is the same for all times of the day.

To enable a context-aware measurement, the relevant contextual factors and their characteristics have to be identified. Contextual factors which are relevant for processes can be grouped into two categories: Environmental factors include e.g., industrial conditions such as high/low regulation, customer characteristics such as standard/special clients, general conditions like hot/cold weather, and characteristics of process instances such as light/heavy weight of parcels to be transported. Internal factors comprise e.g., organizational conditions such as the health situation of employees and the workload within the process like low/high degree of capacity utilization.

The identification of contextual factors is highly dependent on the specific business process. There has to be an assumption which factors might have an impact. However, the analysis should use the above mentioned categories as guidance to evaluate if either existing data fits in these types or additional data is needed (Leyer 2011).

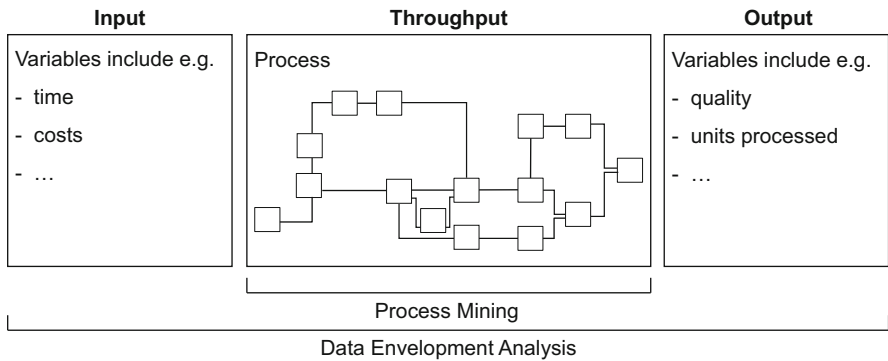
To conduct such an analysis, the characteristics of each contextual factor have to be assigned to every process instance within the process under analysis. The influence of the context can then be determined by using statistical tools like one-way analysis of variance (ANOVA), Mann–Whitney U test or Kruskal–Wallis test (Ruxton and Beauchamp 2008). If statistically significant results are detected, descriptive statistics should be applied to measure the percentile and the average processing time for each characteristic of a contextual factor in order to determine its impact.

## 4 Methodologies for Measurement

As pointed out the development of systems and metrics for process performance analysis is a key issue. However, the use of simple metrics of process analysis is considered critical, as simple approaches fail to take into account the overall performance (Neely et al. 1995). Consequently, it seems more appropriate to use methodologies based on input–output models for performance measurement. Over the years, a number of methodologies for measuring the performance of input–output relations have evolved, at which Frontier Analysis approaches count to be the most popular and precise measurement techniques (Paradi et al. 2004). Within this area several methodologies exist, such as parametric approaches (Stochastic Frontier Approach [SFA], Distribution Free Approach [DFA], Thick Frontier Approach [TFA]) and non-parametric approaches (e.g., Data Envelopment Analysis [DEA], Free Disposable Hull [FDH]). Overall, DEA came up as the most popular methodology for measuring efficiency (Cinca et al. 2002).

DEA is purely based on an input–output model for efficiency measurement. The throughput is considered to be a “blackbox” (Frei and Harker 1999). However,





**Fig. 1** Measurement foci of DEA and Process Mining on a process level

focusing on measurements of the input–output relation is not sufficient because throughput measurements are also needed for root cause analysis of a low performance. Here, Process Mining provides the opportunity to analyze this blackbox in terms of identifying the process model and subsequently the performance of the process (Fig. 1). Process Mining is a methodology which allows evaluating the execution of processes with the help of automatically generated timestamps (van der Aalst 2011). Due to the advantages of the possibility of analyzing large data sets, Process Mining becomes more and more popular.

### 4.1 Data Envelopment Analysis

DEA is a non-parametric, non-stochastic efficiency measurement methodology developed by Charnes et al. (1978). For each object to be measured, termed as Decision Making Unit (DMU), an efficiency score is assessed. This rests upon the derivation of a production frontier determined by the empirically best practice DMU. The distance to the peer object on the best practice frontier determines the efficiency score (Epstein and Henderson 1989). DEA is based on few assumptions only and is especially applicable in cases where input–output relations are analyzed and the production function is unknown (Cooper et al. 2004).

Generally, DEA has been used to detect efficiency on an organizational level (e.g., branches, companies). Since there are always parts of a company, which are more or less efficient than others, measuring on an organizational level might miss opportunities to detect inefficient parts. To overcome this shortcoming, efficiency measurement should be conducted on a process level. Burger and Moormann (2010) present a method to measure efficiency on a process level with Data Envelopment Analysis. In this context, DEA is applied on process instances' level, indicating that each process instance is considered to be a DMU. An instance might be a parcel to be delivered, a loan application to be proceeded etc. Applying

DEA on a process instances' level yields an empirical production function of the process resulting from the best practice process instances. The efficiency score is calculated by the distance to its peer best practice process instance located on the efficient frontier. Aggregating the efficiency scores (e.g. by calculating the mean) delivers an indication of the process efficiency.

## 4.2 Process Mining

As described in Sect. 3.3 several techniques are available to record process execution data. These techniques are usually used manually, i.e., either gathered manually by employees or partly pulled out from an IT system. Unfortunately, these techniques are either imprecise or time-consuming and thus expensive. To avoid these disadvantages, process data should be recorded automatically. Precondition is the implementation of the process within a process-aware information system such as workflow management systems (van der Aalst and Weijters 2004). These systems can generate event logs. Event logs are automatic timestamps documenting order processing. Event logs gathered using techniques of Process Mining enable a relatively fast analysis of processes over a large time span.

Process Mining is a methodology which allows evaluating the execution of processes with the help of automatically generated timestamps. Starting point are customer orders of a process for which the conducted activities at their respective point in time are documented. Beyond these minimum requirements for an analysis it can also be recorded by whom an activity was performed. Every customer order has a unique identification code so that each activity can be assigned properly (van der Aalst 2011). A customer order can thus be described as a sequence of time stamps.

Using the timestamps, the process model can be identified which is a precondition to conduct meaningful measurements. A chosen set of timestamps is analyzed with algorithms to detect the order of performed activities for customer orders (de Medeiros et al. 2007). Using the detected process model, Process Mining techniques can cover three measurement perspectives (van der Aalst et al. 2007):

- *Process perspective:* From this perspective the question is how well customer orders were handled in a process. Thus, the average cycle time, working time, and idle time can be identified. Furthermore, costs can be assigned to evaluate the time needed. The measurement points are determined by the granularity of the process steps implemented in the workflow management system.
- *Organizational perspective:* Within this perspective the performance of the employees working in the process can be measured. Thus, the connection between the resources can be analyzed as well as their degree of capacity utilization.
- *Order specific perspective:* Here, the focus lies on individual customer orders. This perspective is of interest if there is an outlier determining the overall process performance. It can be analyzed for a single order, which route through

the process has been chosen, how the performance has been, and which resources have been working on the respective order.

Despite the benefits it has to be noted that event logs are limited to documented activities monitored by a workflow management system. Activities of employees, like informal phone calls, are not documented.

## 5 Exemplary Application of Process Mining

The example chosen in this section is a loan application process of a medium-sized bank in Germany. In this process corporate loans for small and medium enterprises (SME) without collaterals are handled. Customers of the loan processing are 34 branches of the bank which provide the loans to SMEs. If information or documents are required for handling a SME loan application, the contact takes place through the branches.

When a customer applies for a SME loan at a branch, the loan application is prepared there and forwarded to the loan application process in the bank's back-office. Then, the application is examined by loan clerks. Exceptionally, in the case of high credit amounts, the applications will be first approved by the management board. In a following step it is checked whether the required documents and all needed information have been provided. If documents or information are missing the application is returned to the customer either for supplemental or major revision. If documents and information are complete, there are two possibilities:

- If the loan application has not been approved, it will be done now.
- In the case of special condition agreements, additional processing by a loan officer takes place.

Afterwards, administrative work for all applications will be finished, including possible further inquiries to customers. The latter includes requesting missing signatures of SMEs. If necessary, a post-processing happens as well as a final inspection of the documents leading to archiving the final application.

Data of the loan application process was gathered for a period of 10 months. Within this period 391 loan applications were processed completely, i.e. they were either approved or rejected. The underlying workflow management system documented 7,066 events (timestamps) for all 391 loan applications.

To build the process model, we used the Heuristic Miner of ProM software. This software package makes it possible to analyze event logs via an algorithm, and provides the best results for real-life data (van der Aalst et al. 2004). For the purpose of model building, the data was separated into two halves to ensure an independent and valid procedure. Data from the first half was used to create the model; then the model was tested against the second half. The conformance of the model is 95.0 %, indicating a high representativeness (Rozinat and van der Aalst 2005). Additionally, the model was validated by the process owner of the bank.

**Table 1** Occurrence and working time of activities in the process

| Activities                               | Number of occurrences | Average working time (min) |
|------------------------------------------|-----------------------|----------------------------|
| Pre-checking                             | 619                   | 16                         |
| Processing of incomplete orders          | 71                    | 8                          |
| Preparation of draft for executive board | 4                     | 5                          |
| Approval by executive board              | 3                     | 6                          |
| Approval                                 | 26                    | 18                         |
| Processing approval                      | 8                     | 28                         |
| Preparation of special processing        | 43                    | 12                         |
| Processing of special applications       | 28                    | 17                         |
| Checking of applications                 | 373                   | 11                         |
| Processing of applications               | 173                   | 80                         |
| Processing of reply                      | 1                     | 2                          |
| Re-working                               | 59                    | 24                         |
| Checking of documents                    | 16                    | 55                         |
| Checking of approvals                    | 1                     | 36                         |
| Archiving                                | 377                   | 43                         |

The relevant performance measures of the process are defined as time, costs, and quality. The specific indicators for time are cycle time, working time, waiting time during office hours, and waiting time outside office hours. Based on the timestamps of starting activity and end activity, the respective the cycle time was calculated. The average cycle time of processed loan applications is 22.54 days (MD = 15.25 days, SD = 21.03 days). This average cycle time can be divided into internal idle time outside of business hours (Ø 6.2 days) and internal idle time during business hours (Ø 8.6 days). Therefore, timestamps of customer orders were sorted into the business hours. Additionally, idle time was caused by customers and suppliers but these are not relevant for the performance from an inside perspective. The working time for each loan application itself (Ø 189 min) was identified by calculating the mean time of every activity within the process weighted by the relative occurrence of each activity. Table 1 gives an overview for each activity.

Regarding the costs of processing, the specific indicator was the costs per minute of employees. Further fixed costs did not occur within the process. The work within the process is performed by five groups of employees: Pre-check (1 person, 1.02 €/min), Loan processing (12 persons, 1.02 €/min), Approval (2 persons, 1.13 €/min), Assistant to the Executive Board (1 person, 0.72 €/min) and Executive Board (1 person, 2.04 €/min). These costs can be directly assigned to the individual activities and multiplied with the working time. The average working costs per loan application are 198.45 €.

Quality was measured in terms of back loops of loan applications. If an application is reverted within the process some information has been wrong, unclear, or missing. In such a case, the quality of the loan application is assumed to be insufficient. Back loops can be identified by checking the timestamps for a repeated usage of the same activity. It has to be ensured that an interruption in the same activity, which leads to two timestamps for the same activity in a row, is not

counted. 141 out of the 391 loan applications were affected by back loops leading to a quote of 36.1 % of loan applications with quality issues. For these 141 applications, 178 back loops were detected leading to a back loop rate of 1.26 (SD = 0.34).

Using timestamps it is not only possible to analyze the results for the described performance indicators but also to examine these indicators individually for loan applications. Thus, outliers can be identified and reasons for extreme values can be investigated. Also a correlation between time, costs, and quality can be calculated.

## 6 Conclusion

If a process manager wants to assess opportunities for improving a given business process, he/she should not only rely on a formal depiction and description of the process. He/she also has to analyze the process performance in-depth. Hence, criteria to determine process performance (i.e., indicators, measures, and figures) have to be defined beforehand. Selecting the “right” criteria, however, is a challenge, as there is no agreement in theory or practice on how to do this best.

This chapter has delivered the procedure of process performance measurement as part of the business process management cycle. Within the steps of this procedure relevant approaches have been presented as well as the possibilities to execute the respective steps. The finding is that companies require a process performance measurement system that is tailored specifically to their vision, strategy, and strategic objectives. Additionally, several opportunities to conduct a systematic measurement of indicators have been highlighted. Moreover, two methodologies to measure process performance have been described. Such methodologies allow gathering and identifying the individually relevant measures to conduct a successful management of business processes. Finally, the example of applying Process Mining to a real case indicates the vast opportunities advanced methodologies offer to measure performance of processes.

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